

Cyrus Guided Notes: Stanford AA228/CS238 Decision Making under Uncertainty

Stanford course pack

2026-05-15

Course source

Official source: <https://web.stanford.edu/class/aa228/>

Why this course is in the system. MDPs, POMDPs, planning, risk, and decision-making under uncertainty.

Learning output. Connect it to world models and autonomous-driving decision chapters.

Prerequisite checkpoint

- Probability
- Dynamic programming
- Basic optimization

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned} V^*(s) &= \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right] \\ \pi^*(s) &= \arg \max_a Q^*(s, a) \\ b'(s') &= \eta O(o|s', a) \sum_s P(s'|s, a) b(s) \\ &\quad \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \end{aligned}$$

Visual page

Draw state, action, transition, observation, belief, and value as a decision loop.

GoodNotes pages

- Page ST-DM001: MDP
- Page ST-DM002: POMDP belief update
- Page ST-DM003: risk and planning

Web interaction

A grid-world widget where transition uncertainty changes the best action.

Obsidian and Notion

Layer	Output
Obsidian	Stanford Decision Making -> AA228 CS238
Notion	World-model decision bridge.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.